

IT-SIMPLERA INSTITUTE



Cyber Security Analyst Internship Program

WEEK 02 ASSIGNMENT

Computer Networking Fundamentals
& Network Configuration

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Introduction

Computer networking forms the backbone of modern communication, enabling devices to exchange data locally and across the globe. This report documents the practical and theoretical work completed during Week 2 of the Cyber Security Analyst Internship Program at IT-SIMPLERA Institute, focusing on Computer Networking Fundamentals and Network Configuration.

The report covers five core tasks: identifying network configuration details using the `ipconfig /all` command, executing essential networking diagnostic commands (`ping`, `tracert`, `nslookup`, and `arp`), comparing the roles of routers and switches, contrasting IPv4 and IPv6 addressing, and examining the seven layers of the OSI Model along with their associated protocols and devices. Together, these tasks build a practical and conceptual foundation for further study in network administration and cybersecurity.

Network Information:

IPv4 Address:

An IPv4 address is the unique number that identifies a computer on a network so it can send and receive data.

Example: Just like a house number helps the postman deliver mail to the correct home.

Wireless LAN adapter WiFi:

```
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::7e2:a821:4286:319e%10  
IPv4 Address. . . . . : 192.168.1.5  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::1%10  
                             192.168.1.1
```

MAC Address:

A MAC address is the unique hardware ID of a network card that helps devices identify each other on the same local network.

Example: Like a person's fingerprint—every device has its own unique MAC address.

Ethernet adapter Ethernet:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :  
Description . . . . . : Realtek PCIe GbE Family Controller  
Physical Address. . . . . : B0-0C-D1-EA-B6-4C  
DHCP Enabled. . . . . : Yes  
Autoconfiguration Enabled . . . . : Yes
```

Default Gateway:

A default gateway is the device (usually a router) that sends data from your local network to other networks, such as the internet.

Example: Like the main gate of a school—everyone uses it to go outside the school.

Wireless LAN adapter WiFi:

```
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::7e2:a821:4286:319e%10  
IPv4 Address. . . . . : 192.168.1.5  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::1%10  
                             192.168.1.1
```

Networking Commands

1. ipconfig /all

This command displays detailed network configuration of the computer, including the IPv4 address, MAC address, DNS servers, and Default Gateway.

```
PS C:\Users\malia> ipconfig /all

Windows IP Configuration

Host Name . . . . . : Gurul06
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Realtek PCIe GbE Family Controller
Physical Address. . . . . : B0-0C-D1-EA-B6-4C
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet 3:

Connection-specific DNS Suffix . :
Description . . . . . : VirtualBox Host-Only Ethernet Adapter
Physical Address. . . . . : 0A-00-27-00-00-12
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::a2cd:dbcf:8899:2009%18(Preferred)
IPv4 Address. . . . . : 192.168.56.1(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 722075687
DHCPv6 Client DUID. . . . . : 00-01-00-01-24-06-B7-96-B0-0C-D1-EA-B6-4C
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #3
Physical Address. . . . . : 4A-5F-99-25-2A-A1
```

2. ping google.com

This command checks whether the computer can communicate with Google's server and measures the response time.

```
PS C:\Users\malia> ping google.com

Pinging google.com [142.250.202.174] with 32 bytes of data:
Reply from 142.250.202.174: bytes=32 time=38ms TTL=113
Reply from 142.250.202.174: bytes=32 time=37ms TTL=113
Reply from 142.250.202.174: bytes=32 time=37ms TTL=113

Ping statistics for 142.250.202.174:
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 37ms, Maximum = 38ms, Average = 37ms
Control-C
```

3. tracert google.com

This command shows the path (hops) that data packets travel from your computer to Google's server.

```
PS C:\Users\malia> tracert google.com

Tracing route to google.com [142.250.202.174]
over a maximum of 30 hops:

  1     1 ms     1 ms     1 ms  MYGROUP [192.168.1.1]
  2     3 ms     1 ms     2 ms  10.102.100.1
  3     5 ms     2 ms     4 ms  cpe-1-120-3-1.vb11.vic.asp.telstra.net [1.120.3.1]
  4     4 ms     4 ms    11 ms  10.174.4.234
  5     6 ms     5 ms     6 ms  10.180.53.129
  6     5 ms     3 ms    69 ms  117.20.23.46
  7    24 ms    23 ms    24 ms  110.93.253.182
  8   542 ms    86 ms   127 ms  110.93.252.246
  9    35 ms    38 ms    35 ms  72.14.209.108
 10    36 ms    35 ms    36 ms
```

4. nslookup google.com

This command finds the IP address of a domain name using a DNS server.

```
PS C:\Users\malia> nslookup google.com
Server:  UnKnown
Address:  192.168.1.1

Non-authoritative answer:
Name:     google.com
Addresses: 2a00:1450:4018:811::200e
          142.250.202.206
```

5. arp -a

This command displays the ARP table, which shows the IP addresses and their corresponding MAC addresses of devices on the local network.

```
PS C:\Users\malia> arp -a

Interface: 192.168.1.5 --- 0xa
  Internet Address      Physical Address      Type
  192.168.1.1           28-23-f5-a5-98-90    dynamic
  192.168.1.255         ff-ff-ff-ff-ff-ff    static
  224.0.0.22            01-00-5e-00-00-16    static
  224.0.0.251          01-00-5e-00-00-fb    static
  224.0.0.252          01-00-5e-00-00-fc    static
  239.255.255.250      01-00-5e-7f-ff-fa    static
  255.255.255.255      ff-ff-ff-ff-ff-ff    static

Interface: 192.168.56.1 --- 0x12
  Internet Address      Physical Address      Type
  192.168.56.255       ff-ff-ff-ff-ff-ff    static
  224.0.0.22           01-00-5e-00-00-16    static
  224.0.0.251          01-00-5e-00-00-fb    static
  224.0.0.252          01-00-5e-00-00-fc    static
  239.255.255.250      01-00-5e-7f-ff-fa    static
```

Router vs. Switch

Feature	Router	Switch
Function	Connects different networks and forwards data between them.	Connects devices within the same local network (LAN).
OSI Layer	Works mainly at Layer 3 (Network Layer) .	Works mainly at Layer 2 (Data Link Layer) .
Address Used	Uses IP addresses to route data.	Uses MAC addresses to forward data.
Main Purpose	Provides internet access and connects multiple networks.	Allows computers, printers, and other devices to communicate in the same network.
Data Forwarding	Sends data to the correct network based on the destination IP address.	Sends data directly to the correct device using its MAC address.
Common Use Cases	Home Wi-Fi routers, office networks, and connecting different networks.	Offices, schools, computer labs, and LAN environments where many devices need to communicate.

IPv4 vs. IPv6

Feature	IPv4	IPv6
Address Length	32-bit address	128-bit address
Address Format	Four decimal numbers separated by dots	Eight groups of hexadecimal numbers separated by colons
Number of Addresses	About 4.3 billion addresses	About 340 undecillion addresses (almost unlimited)
Address Example	192.168.1.10	2001:0db8:85a3:0000:0000:8a2e:0370:7334
Configuration	Can be configured manually or by DHCP	Supports automatic configuration (SLAAC) and DHCPv6
Security	IPsec is optional	IPsec support is built into the protocol design
Broadcast Support	Supports broadcast communication	Does not use broadcast; uses multicast instead

OSI MODEL:

OSI Layer (Top to Bottom)	Primary Function	Common Protocol	Networking Device
7. Application Layer	Provides network services directly to user applications.	HTTP	Proxy Server
6. Presentation Layer	Translates, encrypts, and compresses data.	SSL/TLS	Gateway
5. Session Layer	Establishes, manages, and ends communication sessions.	NetBIOS	Gateway
4. Transport Layer	Ensures reliable data delivery and error checking.	TCP	Firewall
3. Network Layer	Routes data between different networks using IP addresses.	IP (IPv4/IPv6)	Router
2. Data Link Layer	Transfers data between devices on the same local network using MAC addresses.	Ethernet (IEEE 802.3)	Switch
1. Physical Layer	Transmits raw bits over cables or wireless signals.	Ethernet (Physical Standard)	Hub / Repeater

Challenges

A few practical challenges came up while completing this assignment. Interpreting the raw output of commands such as `tracert` and `arp -a` took some care, since each hop and table entry needed to be matched back to the concept it illustrated rather than just captured as a screenshot. Keeping the IPv4 and IPv6 comparisons accurate also required double-checking address formats and terminology (for example, the distinction between multicast and broadcast) against reliable references. Finally, mapping each OSI layer to a single representative protocol and device meant simplifying topics that, in practice, often span more than one layer.

Conclusion

This assignment provided a practical introduction to core networking concepts essential for a cybersecurity analyst. By using commands such as `ipconfig /all`, `ping`, `tracert`, `nslookup`, and `arp -a`, hands-on experience was gained in identifying and diagnosing network configuration details, including IPv4 addresses, MAC addresses, and default gateways. Comparing routers and switches clarified how devices operate at different OSI layers to move traffic within and between networks, while contrasting IPv4 and IPv6 highlighted the ongoing shift toward a larger, more secure addressing scheme. Reviewing the seven layers of the OSI Model tied these individual pieces together into a single framework for understanding how data moves from an application down to the physical medium and back. Overall, the exercise strengthened both technical troubleshooting skills and the conceptual foundation needed for further study in network administration and cybersecurity.

References

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